

NOVAC Internship – Task 5 Daily Documentation

Task: React Frontend Migration, Authentication System, Chatbot Integration, Chunk Management, Semantic Search Interface, and Whisper Voice AI Preparation

Overview

Today's work focused on transforming the existing FastAPI-based frontend into a modern React-powered AI application interface with routing, authentication, semantic retrieval interaction, editable chunk management, and voice-AI preparation.

The project evolved from a static HTML upload interface into a multi-page React application capable of supporting Retrieval-Augmented Generation (RAG)-style workflows through a chatbot-based semantic search system.

The task involved learning and implementing:

- React routing
- frontend-backend API communication
- authentication UI workflows
- semantic search interaction
- editable chunk visualization
- MongoDB update workflows
- Whisper voice-to-text preparation

By the end of the session, the system was successfully able to:

- support React-based multi-page navigation
- authenticate users through a login interface
- hide/show passwords dynamically
- upload documents through chatbot UI
- perform semantic search retrieval
- dynamically display semantic search results
- display stored chunks on a dedicated page
- edit and update chunks directly in MongoDB
- prepare voice-to-text infrastructure using Whisper and ffmpeg

Resources and References Used

- React Documentation
- React Router Documentation
- FastAPI Documentation
- MongoDB Documentation
- Sentence Transformers Documentation
- Whisper Documentation
- ffmpeg Documentation
- JavaScript Fetch API Documentation

Concepts Learned

React Routing

Learned how React applications support multi-page navigation using:

- BrowserRouter
- Routes
- Route

Understood component-based frontend architecture.

Implemented navigation between:

- Login page
- Chatbot page
- Chunks management page

Authentication UI Workflow

Studied how frontend login systems work using:

- controlled input states
- password masking
- conditional navigation
- UI validation workflows

Implemented:

- username field
- password field
- password visibility toggle
- login validation

Semantic Search Interface

Learned how semantic retrieval systems can be integrated into chatbot workflows.

Understood:

- frontend query submission
- backend semantic retrieval APIs
- similarity-based response generation
- semantic embedding retrieval workflows

Implemented semantic search directly inside chatbot UI.

MongoDB Chunk Editing Workflow

Learned how frontend interfaces interact with databases through API endpoints.

Implemented:

- chunk retrieval APIs
- editable textareas
- chunk update requests
- dynamic frontend rendering

Understood how MongoDB updates occur through backend API communication.

Whisper Voice-to-Text Preparation

Studied OpenAI Whisper as an open-source speech recognition model.

Installed:

- openai-whisper
- ffmpeg
- PyTorch dependencies

Learned how audio transcription pipelines work using:

- microphone input
- audio processing
- speech-to-text conversion

Prepared the backend for future voice-enabled chatbot interaction.

Libraries, Modules, and Components Learned

React.js: Used for frontend component development and UI rendering.

React Router: Used for multi-page navigation and route management.

FastAPI: Used for backend API development and request handling.

MongoDB: Used for storing semantic chunks and document data.

Sentence Transformers: Used for semantic embedding generation and retrieval.

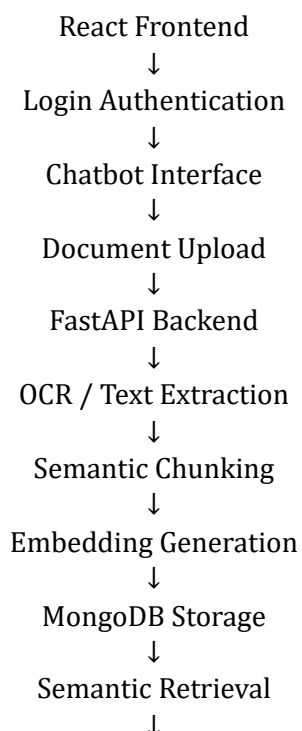
Whisper: Used for speech-to-text preparation workflows.

ffmpeg: Used for audio processing support required by Whisper.

Practical Implementation

- Migrated frontend from HTML to React
- Configured React routing system
- Created Login.jsx page
- Created Chatbot.jsx page
- Created Chunks.jsx page
- Added password visibility toggle
- Added login validation workflow
- Connected React frontend with FastAPI backend
- Integrated semantic search into chatbot interface
- Added chunk retrieval page
- Added editable chunk visualization
- Implemented MongoDB chunk update workflow
- Added frontend API fetch requests
- Improved frontend page navigation
- Installed Whisper dependencies
- Installed ffmpeg for audio processing
- Prepared speech-to-text integration workflow

Architecture Understood



Challenges Faced

React Routing Errors

Encountered issues related to:

- missing route definitions
- incorrect component imports
- blank page rendering

Resolved by properly configuring:

- BrowserRouter
- Routes
- Route paths

Frontend-Backend Integration Issues

Encountered:

- API response mismatches
- request formatting issues
- upload validation errors

Resolved through endpoint restructuring and fetch request corrections.

Chunk Display Issues

Faced problems where:

- chunks were not rendering
- routes displayed blank screens
- MongoDB data was not displayed correctly

Resolved by:

- fixing React routes
- adding chunk retrieval endpoints
- updating frontend rendering logic

Dependency Installation Issues

Faced temporary installation and internet resolution issues while installing:

- openai-whisper
- ffmpeg dependencies

Successfully resolved and completed installation.

Conclusion

Today's work significantly upgraded the project from a simple AI ingestion pipeline into a foundational RAG-ready semantic document intelligence system.

The system now supports:

- authentication workflows
- React-based frontend architecture
- chatbot-driven semantic retrieval
- editable chunk management
- MongoDB synchronization
- semantic search interaction
- voice-AI preparation workflows

The concepts learned today form the foundation of:

- AI-powered document assistants
- enterprise RAG systems
- semantic chatbot platforms
- multimodal AI interfaces
- voice-enabled AI applications

Prepared By

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